

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. (CL) Examination

November 2012

CL 210 Hydrology and Hydraulics

Date: 07.11.2012, Wednesday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) What are different types of precipitation? Explain any one. [2]
- (b) Test the consistency of the 22 years of data of annual precipitation measured at station 'A'. Rainfall data for the station 'A' as well as the average annual rainfall measured at a group of eight neighboring stations located in a meteorologically homogeneous region are given below: [5]

Year	1986	1987	1988	1989	1990	1991	1992	1993	1994	1995	1996
Station A (cm)	177	144	178	162	194	168	196	144	160	196	141
Neighboring Eight Stations Average (cm)	143	132	146	147	161	155	152	117	128	193	156
Year	1997	1998	1999	2000	2001	2002	2003	2004	2005	2006	2007
Station A (cm)	158	145	132	95	148	142	140	130	137	130	163
Neighboring Eight Stations Average (cm)	164	155	143	115	135	163	135	143	130	146	161

- Q - 2 (a) (i) Explain the terms: Φ -index and ω -index. [2]
- (ii) The average rainfall over a basin of area of 50 ha. during a storm were noted and listed in table given below. If the volume of runoff from this storm was measured as 25,000 m³, determine Φ -index for the basin. [3]

Time (hr)	0	1	2	3	4	5	6	7
Rainfall (mm)	0.0	6.0	11.0	34.0	28.0	12.0	6.0	0.0

- (b) What are different methods for flood frequency analysis estimation of flood discharge? Discuss any one in detail. [5]
- (c) (i) Distinguish between fan shaped and fern leaf catchment. [2]
- (ii) List out different methods for reducing evaporation from a water body. [2]

OR

- Q - 2 (a) (i) Explain the usefulness of the rainfall data analysis. [2]
- (ii) What is Unit Hydrograph (UH)? Write the assumptions made in derivation of equation of it. Enlist applications of UH. [4]

- (b) A catchment area of a basin having a shape of regular hexagon of side 40 km. Six rain gauges are located at end of the hexagon side. The rainfall recorded are 20, 25, 30, 35, 18 and 32 cm. Using Thiessen's polygon method, compute the average precipitation over the basin. Also calculate the average rainfall in the basin using arithmetic mean method. [3]
- (c) Find out the ordinates of a storm hydrograph resulting from a 3-hr storm with rainfall of 3.0, 4.5 and 1.5 cm during subsequent 3-hr intervals. The ordinates of UH are given in table below: [5]

Time (hrs)	0	3	6	9	12	15	18	21	24	27	30	33	36
UH Ordinate (cumecs)	0	90	200	350	450	350	260	190	130	80	45	20	0

Assume an initial loss is 5 mm, infiltration index 5 mm/hr and base flow 20 cumecs.

- Q - 3 (a) What is meant by interference between tubewells? What is an ideal distance between two adjacent tubewells? [2]
- (b) Define the terms Artesian well, Specific yield, Storage coefficient and Transmissibility. [4]
- (c) Two tubewells each of 20 cm diameter are spaced at 100 m distance. Both the wells penetrate fully a confined aquifer of 12 m thickness. Calculate the discharge if only one well is discharging under a depression head of 3 m. What will be the percentage decrease in the discharge of the well, if both the wells are discharging under the depression head of 3 m? Take radius of influence for each well equal to 250 m and coefficient of permeability of aquifer as 60 m/day. [8]

OR

- Q - 3 (a) Write short notes on following sub-surface ground water investigation techniques: [8]
- (i) Electrical resistivity (ii) Radioactive logging
- (b) Explain Dupuit's theory for the steady radial flow to a well. What are the assumptions made in the theory? Also obtain an expression for the discharge from a well fully penetrating a confined aquifer. [6]

SECTION – II

- Q - 4 Explain the process of ground water recharge in detail. What are the needs to recharge a ground? Classify the methods of artificial recharge. [7]
- Q - 5 (a) Define following terms: [4]
- (i) Aquifer (ii) Aquiclude (iii) Aquifuge (iv) Aquitard
- (b) An aquifer has 27 m thickness, 33% porosity and bulk modulus of compression as 10^8 N/m^2 . Estimate the storage coefficient of an aquifer. Assume, $K_w = 2.1 \text{ GN/m}^2$. [4]

- (c) Define Conjunctive Use. What are the needs of it? [4]
 (d) Write the advantages and disadvantages of roof Rain Water Harvesting system. [2]

OR

Q - 5 (a) An aquifer of aerial extent of 110 km^2 is overlain by four strata as given below: [8]

Strata	Thickness (m)	Kx (m/day)	Ky (m/day)
1 (Top)	2	3.2	0.54
2 (Middle)	3	1.6	0.63
3 (Middle)	2	1.3	0.065
4 (Bottom)	1	0.04	0.004

- (i) If a 3-hr storm occurs producing a total rainfall 125 mm, estimate the recharge into the aquifer, assuming that the piezometric surface in the aquifer is at the bottom of layer and that all layers are saturated.
 (ii) If all three layers are underlain by an impermeable strata instead of the aquifer, estimate the lateral flow per unit width through layers, assuming the layer dip by 0.20 %.

- (b) Explain the laboratory methods of measurement of permeability in detail. [4]
 (c) Write an approximate demand of water for rural water supply, live stock and industry. [2]

Q - 6 (a) Derive the Gyben-Herzberg relation for salt water-fresh water interface and explain concept of slope & shape of it. [7]

- (b) What is ground water pollution? Explain the pollution mechanism with neat diagram. [7]
 Also discuss factors affecting ground water pollution.

OR

Q - 6 (a) Which prevention measures should be taken to reduce the salt water intrusion? Discuss the methods used for controlling it with neat sketches. [7]

- (b) Discuss the sources and nature of pollution from industrial, agricultural, sewage and nuclear wastes. [7]
